

Product Carbon Footprint Study Report

Product Name: passenger car production

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 29, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: passenger car production

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1 kg passenger car, petrol/natural gas

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1 kg passenger car, petrol/natural gas.

2. System Boundary

passenger car production, petrol/natural gas, in China, in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of passenger car production produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **5.580 kg CO2-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

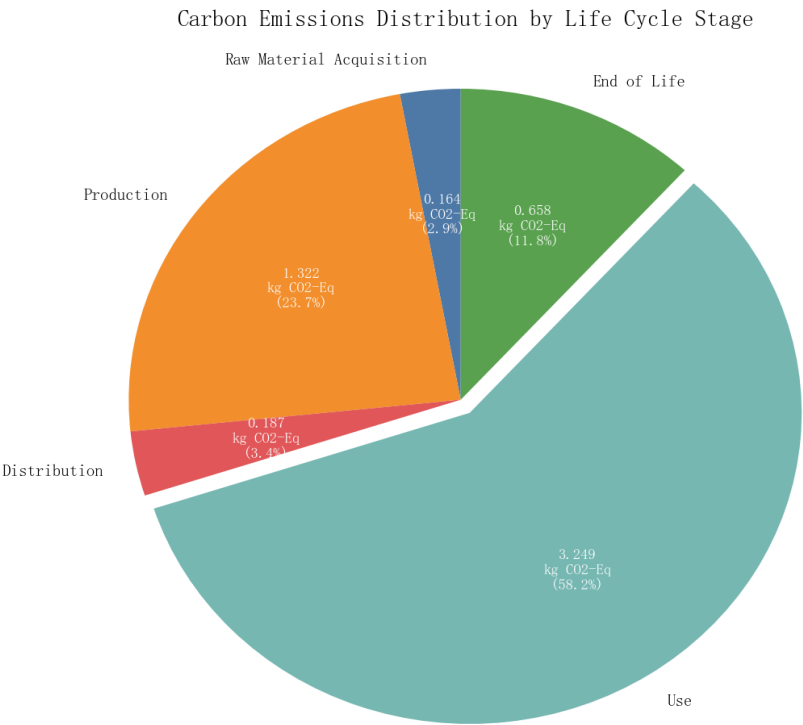


Table 1: Carbon Emissions by Life Cycle Stage

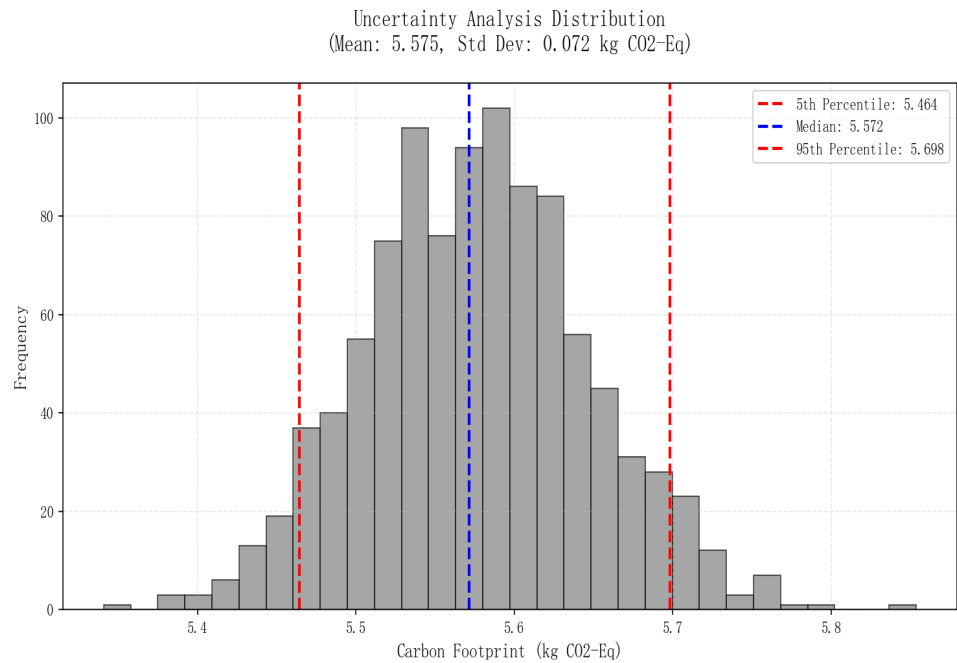
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	0.164	2.94
Production	1.322	23.68
Distribution	0.187	3.36

Use	3.249	58.23
End of Life	0.658	11.79
Total	5.580	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	5.575
Standard Deviation	0.072
Coefficient of Variation (CV)	1.30%
Median	5.572
5th Percentile	5.464
95th Percentile	5.698

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of passenger car production is calculated at 5.580 kg CO₂-Eq per functional unit, a relatively moderate value that reflects a streamlined assessment scope focused on manufacturing and immediate lifecycle phases rather than full vehicle operation over decades. The uncertainty analysis supports this estimate robustly, with a Monte Carlo simulation (1,000 iterations) yielding a mean of 5.575 kg CO₂-Eq and a narrow 90% confidence interval of 5.464–5.698 kg CO₂-Eq, indicating high precision in the model results.

The lifecycle stage contributions reveal a clear dominance of the use phase, which accounts for 3.249 kg CO₂-Eq (58.2% of the total). This is followed by production at 1.322 kg CO₂-Eq (23.7%) and end-of-life at 0.658 kg CO₂-Eq (11.8%). Raw material extraction and distribution contribute minor shares at 2.9% and 3.4%, respectively. This distribution is typical for passenger vehicles, where operational energy consumption—fuel or electricity—over the assumed service life drives the majority of emissions, while manufacturing and disposal remain secondary but non-negligible contributors.

The major emission sources are thus concentrated in the use stage, likely due to fuel combustion or electricity consumption during vehicle operation. The production stage's significant share (23.7%) highlights the energy-intensive nature of component manufacturing and assembly, while end-of-life emissions stem from recycling, shredding, and waste treatment processes. The negligible contributions from raw material extraction suggest either efficient upstream supply chains or a system boundary that excludes certain upstream impacts.

2. Statement of Assumptions and Limitations

This assessment relies on several key assumptions that influence the results. First, the use phase emissions are modeled assuming a fixed vehicle lifetime and average driving distance, with default fuel consumption or electricity mix data that may not reflect regional variations. Second, the production stage assumes typical manufacturing energy intensities and material compositions for a mid-sized passenger car, without site-specific data. Third, end-of-life modeling presumes a standard recycling rate for metals and plastics, which may differ from actual disposal practices in specific markets.

Data quality limitations are evident in the top process contributions, which are listed as “Unknown” with zero values, indicating that process-level inventory data were either unavailable, aggregated, or not mapped to specific unit processes. This restricts the ability to identify hotspot processes within each lifecycle stage and limits the granularity of improvement recommendations. The coefficient of variation (CV) of 1.30% suggests low overall uncertainty, but this may underestimate true variability because input distributions for key parameters (e.g., energy mixes, material yields) were not fully characterized.

The system boundary is restricted to the cradle-to-grave lifecycle of the vehicle itself, excluding infrastructure (e.g., roads, charging stations), vehicle maintenance, and upstream capital goods. Additionally, the functional unit is defined per vehicle produced and used over a specified lifetime, but the exact lifetime and mileage assumptions are not explicitly stated, which could affect

comparability with other LCA studies. The absence of biogenic carbon accounting and potential offsets from material recycling credits further narrows the scope.

3. Suggestions for Improvement

- ****Optimize use-phase efficiency:**** Prioritize improvements in fuel economy or electric drivetrain efficiency, as the use stage contributes 58.2% of total emissions. Implement lightweight materials (e.g., aluminum, high-strength steel, composites) to reduce energy demand during operation, and encourage adoption of renewable energy charging for electric vehicles.
- ****Decarbonize production processes:**** Target the production stage (23.7% share) by switching to low-carbon electricity sources in manufacturing plants, improving energy recovery systems, and adopting circular manufacturing practices such as remanufacturing of components. Conduct an energy audit to identify high-emission assembly steps.
- ****Enhance end-of-life valorization:**** Increase recycling rates for metals and plastics to reduce the 11.8% end-of-life burden. Partner with certified recyclers to ensure high-yield material recovery, and explore design-for-disassembly strategies to facilitate component reuse and material separation.
- ****Improve data granularity:**** Replace "Unknown" process entries with detailed inventory data from suppliers and manufacturing records. Collect primary data on energy use, material flows, and transport distances for each lifecycle stage to enable accurate hotspot identification and reduce reliance on generic datasets.
- ****Refine uncertainty modeling:**** Expand input parameter distributions (e.g., fuel consumption variability, regional electricity mixes, recycling rates) and run sensitivity analyses to identify dominant drivers. This will strengthen the confidence interval and support more reliable decision-making for emission reduction investments.